

Online cache update

Historical cache $H_{<i}$



Current chunk x_i



Current KV



Gather KV with m_i



Updated cache $\rightarrow$ next chunk



History

Current chunk

Current KV

Retained KV

Predict

Cache KV

Retention mask m_i

DeCoPrune: denoising-consistency mask

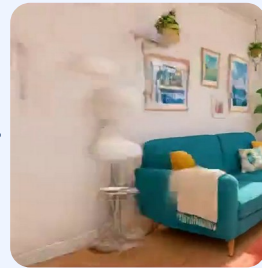
1 Compare clean predictions



Context



Noise

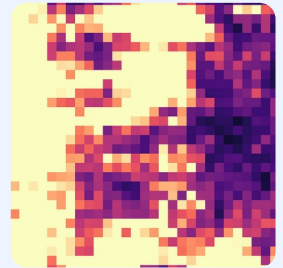


Probe: $\hat{x}_{0,i}(\tau^*)$



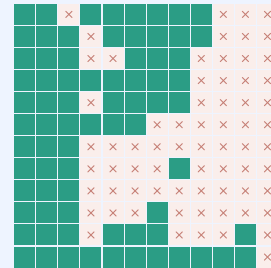
Final: x_i^{final}

2 Measure discrepancy



$$\ell_{i,p} = \|\hat{x}_{0,i,p}(\tau^*) - x_{i,p}^{\text{final}}\|_2^2 / d$$

3 Threshold into a retention mask



Low: prune

High: retain

$$m_{i,p} = 1[\ell_{i,p} > \gamma], \quad 1 = \text{retain}$$